

Autodesk® Scaleform®

Scaleform 4.4

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Autodesk® Scaleform® 4.4

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1 소개

Scaleform 4.0 () Scaleform SDK
Scaleform 4.0 API
Scaleform

2 멀티 쓰레딩 렌더러 변경 내용

- 4.0 API Scaleform 3.x Scaleform
1. Renderer2D
 2. GFxLoader
 3. - (thread-safe)
 4. MovieDisplayHandle
Renderer2D::Display
 5. GFx::Movie
 6. API
Scaleform 3.X GRenderer Render::HAL

2.1

```
Scaleform                                Render::Renderer2D    Render::HAL
      . Renderer2D                                     Display
      . Render::HAL    Renderer2D
      ,                Render::D3D9::HAL    Render::PS3::HAL
```

```
D3DPRESENT_PARAMETERS    PresentParams;
IDirect3DDevice9*         pDevice = ...;
ThreadId                  renderThreadId = Scaleform::GetCurrentThreadId();
Render::ThreadCommandQueue *commandQueue = ...;
Ptr<Render::D3D9::HAL>    pHal;
Ptr<Render::Renderer2D>  pRenderer;

pHal = *SF_NEW Render::D3D9::HAL(commandQueue);

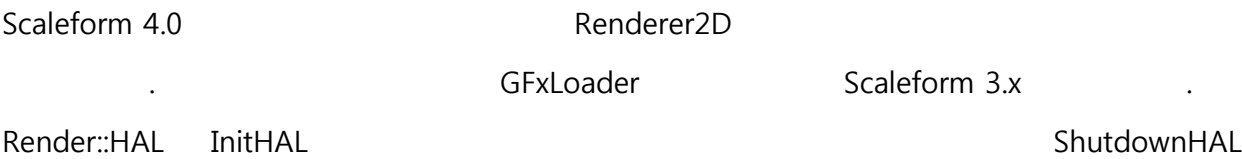
if (!pHal->InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
                                                renderThreadId)))
{
    return false;
}
pRenderer = *SF_NEW Render::Renderer2D(pHal);
```

```
renderThreadId      pDevice      PresentParams      , HALInitParams
```

```
. ThreadCommandQueue
```

Renderer2D HAL - (thread-safe)

2.2



Render::HAL

HAL

```
MeshCacheParams mcp;
mcp.MemReserve      = 1024 * 1024 * 3;
mcp.MemLimit        = 1024 * 1024 * 12;
mcp.MemGranularity  = 1024 * 1024 * 3;
mcp.LRUTailSize     = 1024 * 1024 * 4;
mcp.StagingBufferSize = 1024 * 1024 * 2;

pRenderer->GetMeshCacheConfig()->SetParams(mcp);
```

, MemReserve MemLimit



PC

```
GlyphCacheParams gcp;
gcp.TextureWidth  = 1024;
gcp.TextureHeight = 1024;
gcp.NumTextures   = 1;
gcp.MaxSlotHeight = 48;
```



```
pRenderer->GetGlyphCacheConfig()->SetParams(gcp);
```

```

                                Scaleform 3.3
                                . TextureWidth, TextureHeight
NumTextures                                . MaxSlotHeight
                                .
                                . InitHAL
InitHAL                                . InitHAL
SetParams                                .

```

2.3

```

Scaleform 4.0                                GFX::ImageCreator
Render::TextureManager                                .
ImageCreator                                GFX::Loader
                                .

```

```

GFX::Loader loader;
...
SF::Ptr<GFX::ImageCreator> pimageCreator =
    *new GFX::ImageCreator(pRenderThread->GetTextureManager());
loader.SetImageCreator(pimageCreator);

```

```

ImageCreator                                TextureManager
TextureManager                                Render::HAL::GetTextureManager()
                                .
                                . GetTextureManager
                                .

```

Render::HAL , CreateTexture TextureManager
 - (thread-safe) .
 .
 • D3DCREATE_MULTITHREADED
 D3D9 - (thread-safe) .
 • ThreadCommandQueue ,
 Render::HAL ..
D3D
ThreadCommandQueue .
 2 . to
 Platform::RenderHALThread .

2.4

. advance
 advance/ "draw frame" -
 . Scaleform 4.0 APIs Gfx::MovieDisplayHandle
 . Gfx::Movie -
 Display ..
 .
 1. . Gfx::Movie Gfx::MovieDef::CreateInstance
 , .
 2. . Advance
 ActionScript . Advance
 Invoke/DirectAccess API .
 . Advance Scaleform
 .

3.

MovieDisplayHandle

```
//-----  
// 1.  
  
Ptr<Gfx::Movie>      pMovie = ...;  
Gfx::MovieDisplayHandle hMovieDisplay;  
  
//  
//  
pMovie->SetViewport(width, height, 0,0, width, height);  
hMovieDisplay = pMovie->GetDisplayHandle();  
  
//                                     . Scaleform Player  
//  
pRenderThread->AddDisplayHandle(hMovieDisplay);  
  
//-----  
// 2.  
  
//  
// ExternalInterface  
float deltaT = ...;  
pMovie->HandleEvent(...);  
pMovie->Advance(deltaT);  
  
//  
//  
pRenderThread->WaitForOutstandingDrawFrame();  
pRenderThread->DrawFrame();  
  
//-----  
// 3.           -           DrawFrame  
Ptr<Render::Renderer2D> pRenderer = ...;  
  
pRenderer->BeginFrame();  
bool hasData = hMovieDisplay.NextCapture(pRenderer->GetContextNotify());  
if (hasData)  
    pRenderer->Display(hMovieDisplay);  
pRenderer->EndFrame();
```

- WaitForOutstandingFrame – CPU
 .
 .
 , Advance
 .
- MovieDisplay.NextCapture – Advance " "
 .
 false
 .

2.5

GFx::Movie Release .

Render::ThreadCommandQueue Movie .

ThreadCommandQueue .

, advance

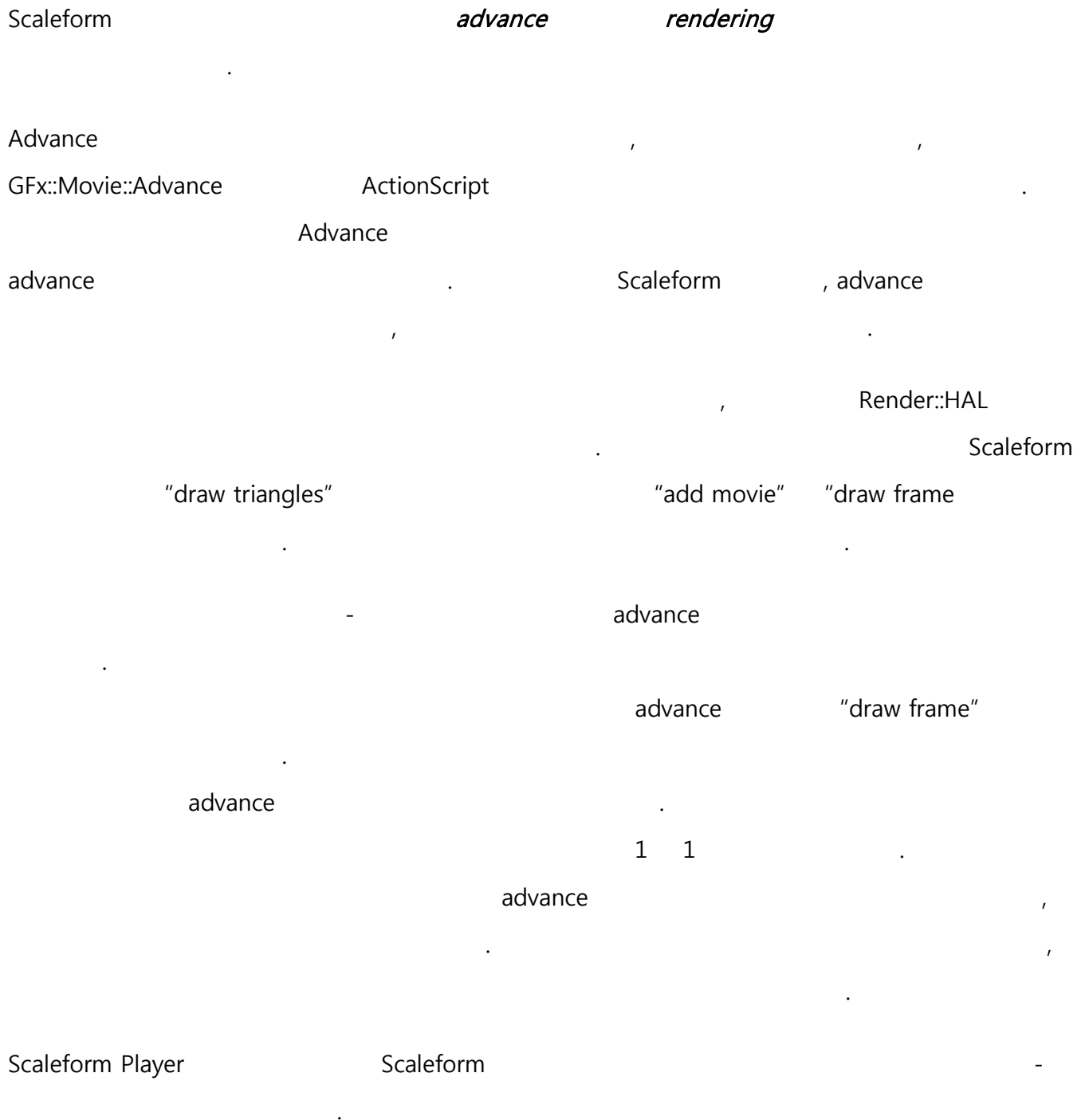
Movie::ShutdownRendering(false)

Movie::IsShutdownRenderingComplete .

, NextCapture false . advance

.

3 멀티 쓰레드 렌더링 개념



3.1

Advance

/

Scaleform ,

GFx::Movie::Advance

GFx::Movie::Capture

PN5Td(tC)19.0012Tw 6(tu(5Tr)Tj0.37 0 Td0891110761
565(0012/095M12/11/71

3.2

Render::Renderer2D::Display

.

Render::Renderer2D::Display

.

Display

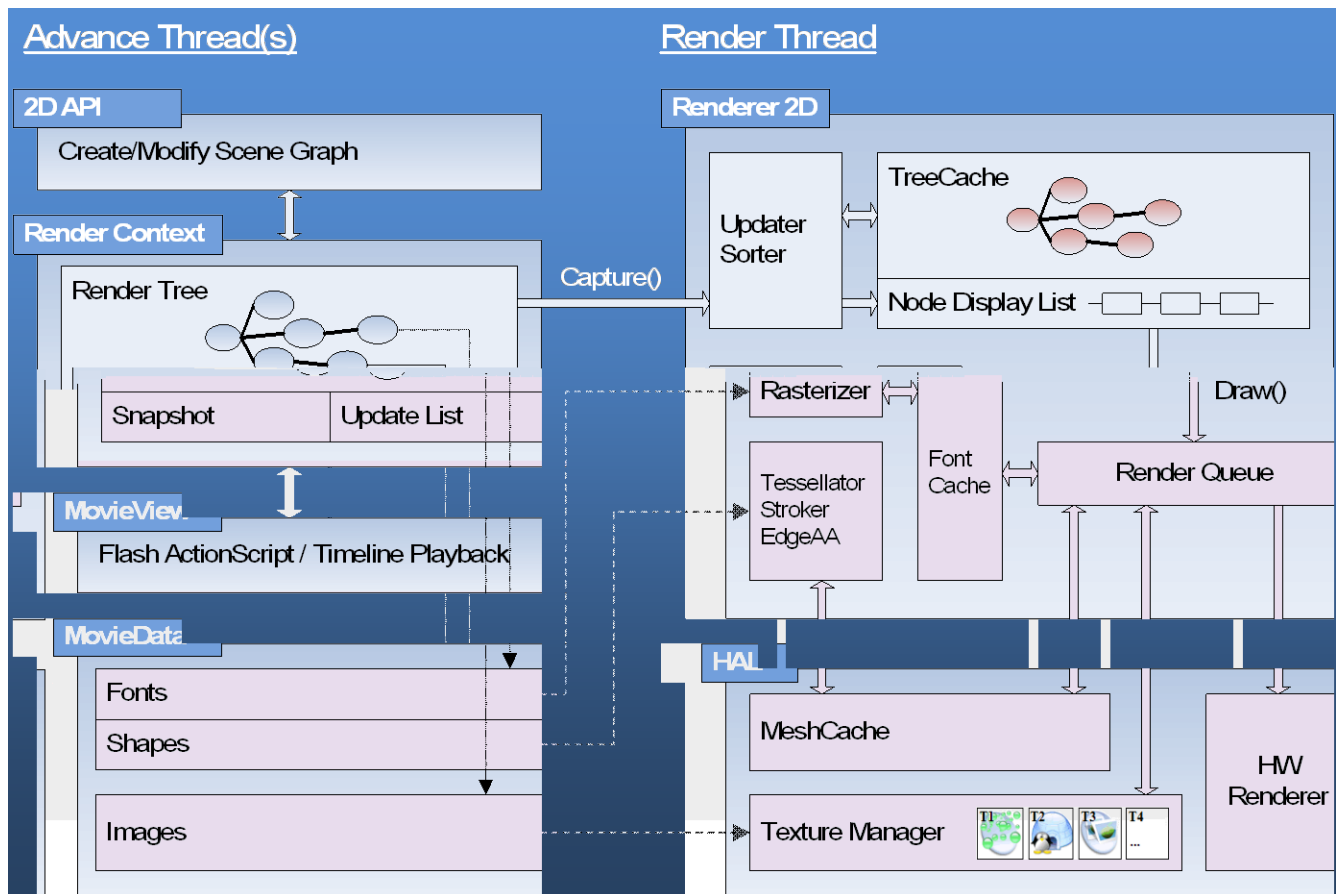
.

. Scaleform

.

4 렌더러 설계:

Scaleform 4.0



1:

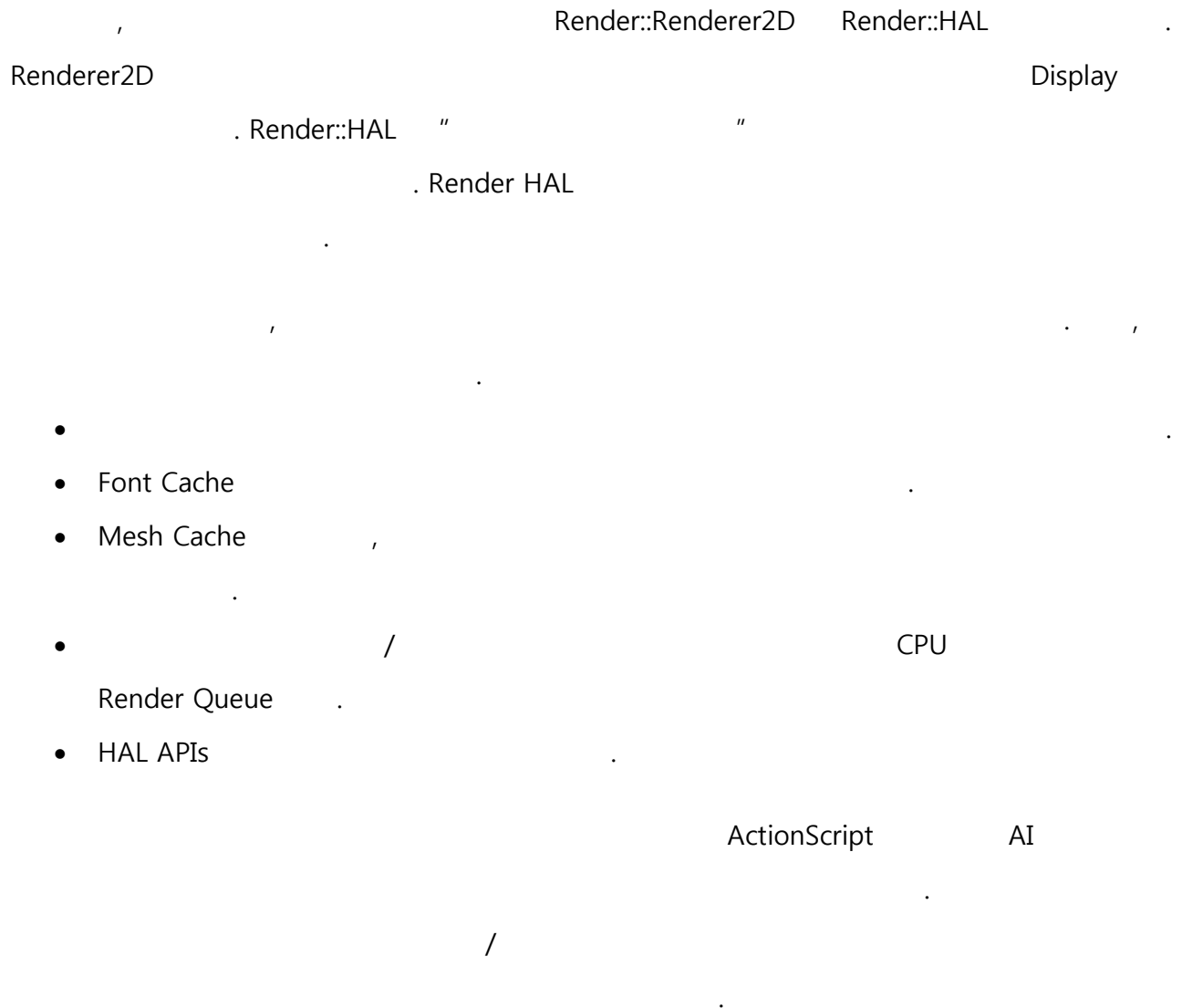
MovieView MovieData Scaleform

. Render Context Render Tree

2D

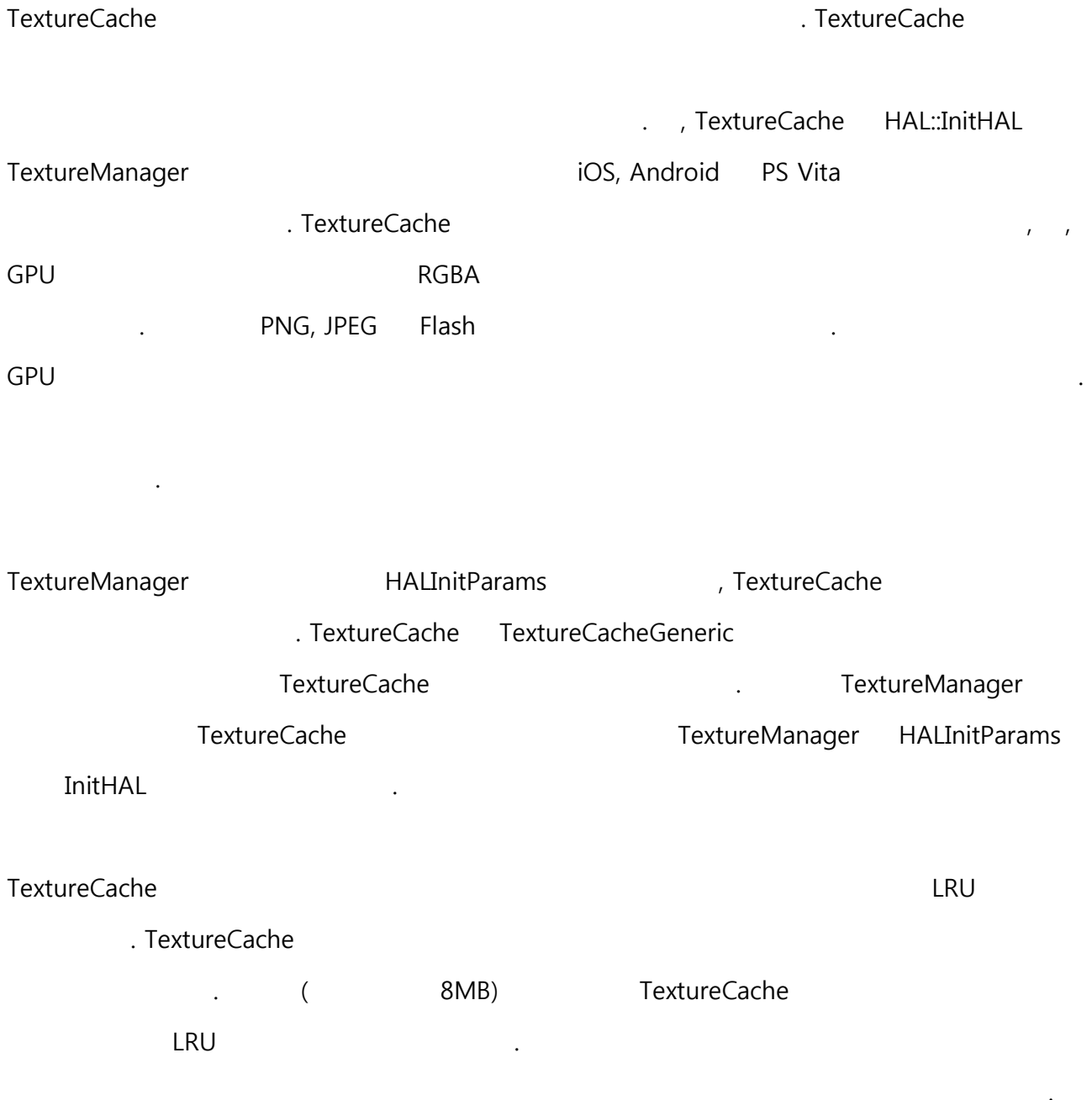
. Scaleform

Movie



5 렌더링 하위 시스템 설명

5.1 TextureCache



TextureCache SetLimitSize

6 사용자 정의 데이터로 렌더링

Scaleform

6.1

AS2
"renderString" "renderFloat" AS3
"setRenderFloat" ActionScript "setRenderString"
AS2/AS3
Movie clip / float
UserDataState::Data the HAL::PushUserData Render::HAL UserDataStack
movie clip / Float Movie clip
UserDataState::Data HAL::PopUserData

6.2

movie clip HAL
HAL
2D
'm' movie clip AS3 AS3 SWF

```
import scaleform.gfx.*;
DisplayObjectEx.setRendererString(m, "Abc");
DisplayObjectEx.setRendererFloat(m, 17.1717);
```

4.1

Src/Render/ShaderData.xml

```
<ShaderFeature id="Position">
  <ShaderFeatureFlavor id="Position2d" hide="true"/>
  <ShaderFeatureFlavor id="Position3d"/>
  <ShaderFeatureFlavor id="Position2dOffset"/>
</ShaderFeature>
```

"Position2dOffset"

snippet

"Position2dOffset" snippet

```
<ShaderSource id="Position2dOffset" pipeline="Vertex">
  Position2d(
    attribute float4 pos      : POSITION,
    varying   float4 vpos     : POSITION,
    uniform   float4 mvp[2],
    uniform   float  poffset)
  {
    vpos = float4(0,0,0,1);
    vpos.x = dot(pos, mvp[0]) + poffset;
    vpos.y = dot(pos, mvp[1]) + poffset;
  }
</ShaderSource>
```

snippet x y 'poffset' (uniform)

"Position2d" snippet

D3D9_Shader.cpp ShaderInterface::SetStaticShader

```
// See if we have user data on the stack.
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
```

```

        if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
            data->RendererString.CompareNoCase("abc") == 0)
        {
            // Modify the shader we use, if we find the matching user data we were
expecting.
            vshader = (VertexShaderDesc::ShaderType)((int)vshader +
VertexShaderDesc::VS_base_Position2dOffset);
            shader = (FragShaderDesc::ShaderType) ((int)shader +
FragShaderDesc::FS_base_Position2dOffset);
        }
    }
}

```

```

        HAL    UserDataStack    ("abc"        float        )
                                .        Position2dOffset
                                .
                                ShaderData.xml
D3D9_ShaderDescs.h
                                .
                                'poffset'    (uniform)
                                .
                                ShaderInterface::Finish
                                .

```

```

// See if we have user data on the stack.
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
        data->RendererString.CompareNoCase("abc") == 0)
    {
        // Add the poffset uniform into the uniform data.
        for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
        {
            float poffset = data->RendererFloat / 256.0f;
            SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
        }
    }
}
}

```

```

        'meshCount'
draw(                                     ) meshCount
        movie clip 'm'

```

6.3

```

HAL      Scaleform
        .
        .
        .
Scaleform      Scaleform
        .
        AS3 "disableBatching"
        AS2 "disableBatching"
        . AS2/AS3
        .

```

7 GFxShaderMaker

7.1

```
Scaleform 4.1      GfxShaderMaker    Scaleform
                  (Render_D3D9)          ShaderData.xml
                  .                      ,
                  .                      .
                  .                      D3D9
```

```
Src/Render/D3D9_ShaderDescs.h
Src/Render/D3D9_ShaderDescs.cpp
Src/Render/D3D9_ShaderBinary.cpp
```

7.2 API HAL

7.2.1 D3D9

```
D3D9 HAL ShaderModel 2.0 ShaderModel 3.0
HAL
GFXShaderMaker '-shadermodel'
```

GFxShaderMaker.exe -platform D3D1x -shadermodel SM30

ShaderModel 2.0

. ShaderModel 2.0

InitHAL

, .

7.2.2 D3D1x

D3D1x HAL

3

. (D3D11 D3D10.1)

FEATURE_LEVEL_10_0

FEATURE_LEVEL_9_3

FEATURE_LEVEL_9_1

, HAL .

. GfxShaderMaker '-featurelevel' ,

. :

GfxShaderMaker.exe -platform D3D1x -featurelevel

FEATURE_LEVEL_10_0,FEATURE_LEVEL_9_1

FEATURE_LEVEL9_3

. FEATURE_LEVEL_9_3

InitHAL

, (FEATURE_LEVEL_9_1)

.